



ENGG100 - Practical 5

Engineering Computing and Analysis (University of Wollongong)



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ENGG100 – Engineering Computing and Analysis

Practical 5 – Week 5

In this Practical class, you will work through parts of the third and seventh chapters of the subject textbook “Essential MATLAB for engineers and scientists” (7th Edition) by Brian H. Hahn and Daniel T. Valentine, 2019.

STEP 1 – Access the online textbook through the library’s portal.

- 1.1 Access the textbook using the same process as Practical 1. Access the PDF version of Chapters 3 and 7. Note the objectives of each chapter are highlighted in orange.

STEP 2 – Section 3.2 – Programming MATLAB functions

- 2.1 Read and perform the tasks in Section 3.2.1 – Inline objects: harmonic oscillators and Section 3.2.2 – MATLAB function: $y = f(x)$. If you need a reminder of the quadratic function from last week, see **Practical 4, Step 2.2**. Check the output of your quadratic function for various inputs.
- 2.2 Practice creating functions by performing Chapter Exercises 3.2, 3.4, 3.6. In each case, develop a structure plan (if not already given) or draw a flowchart, then implement the solution as a function. Be sure to check your functions with various inputs. Show your tutor the output for this section.

STEP 3 – Chapter 7 – Function M-files

- 3.1 Read through the introduction to Chapter 7 and the example in Section 7.1. There is no need to perform the tasks, however, note that multiple files can be saved (eg. “**f.m**” and “**df.m**”) and called from separate script files (eg. “**newtgen.m**”) or from the command line.
- 3.2 Create the file “**stats.m**” in Section 7.2 – Basic Rules and perform the tests. Then read through the general form of a function and check the output of all examples in this section. Skip Sections 7.2.1-7.2.4.
- 3.3 Read through Section 7.3 – Function Handles, Section 7.4 – Command/Function Duality and Section 7.5 – Function Name Resolution.
- 3.4 For additional practice using functions, complete the Chapter Exercises 7.3, 7.4, 7.7. In each case, develop a structure plan or draw a flowchart, then implement the solution as a function m-file. Discuss possible solution approaches with your peers.